

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1353

Sheridan Alte 138 kV
December 31, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1353. The Interconnection Customer proposes a 175.9 MW solar plus storage facility interconnecting at the Sheridan Alte 138 kV substation in the Virginia area. The request was received on December 09, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 337 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$19,020,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Hondo-Cloverdale 138kV
Contingency: Blue Ridge-Sheridan Alte 138kV
Loading: 171.3% (Rating: 195 MVA)
DFAX: 100.0%
Pre-project loading: 81.0%
2. Facility: Mt. Union-Cloverdale 138kV
Contingency: Blue Ridge-Sheridan Alte 138kV L
Loading: 155.2% (Rating: 192 MVA)
DFAX: 100.0%
Pre-project loading: 63.7%
3. Facility: Blue Ridge-Sheridan Alte 138kV L
Contingency: Cloverdale Substation-Hondo 138kV
Loading: 153.3% (Rating: 218 MVA)
DFAX: 100.0%
Pre-project loading: 72.5%
4. Facility: Mt. Union-Sheridan Alte 138kV
Contingency: Sheridan Alte Substation-Blue Ridge 138kV line
Loading: 150.4% (Rating: 205 MVA)
DFAX: 100.0%
Pre-project loading: 64.6%
5. Facility: Vc Porterdale Hydroelectric-Neely 138kV L
Contingency: Cloverdale-Hondo 138kV
Loading: 106.1% (Rating: 277 MVA)
DFAX: 52.3%
Pre-project loading: 72.8%

6. Facility: Blue Ridge-Neely Substation 138kV
Contingency: Cloverdale-Hondo 138kV
Loading: 105.3% (Rating: 295 MVA)
DFAX: 52.3%
Pre-project loading: 74.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hondo-Cloverdale 138kV	Rebuild line with higher-capacity conductor	\$4,968,000	44% (\$2,209,000)
Cloverdale-Mt. Union 138kV	Rebuild line with higher-capacity conductor	\$6,292,000	54% (\$3,413,000)
Blue Ridge-Sheridan Alte 138kV L	Rebuild line with higher-capacity conductor	\$17,941,000	44% (\$7,977,000)
Mt. Union-Sheridan Alte 138kV	Rebuild line with higher-capacity conductor	\$8,679,000	54% (\$4,707,000)
Vc Porterdale Hydroelectric-Neely 138kV Line	Reconductor line	\$746,000	27% (\$201,000)
Blue Ridge Substation-Neely 138kV L	Reconductor line	\$1,909,000	27% (\$513,000)
	Total allocated to project		\$19,020,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.